

WEEK – 2

```
mysql> CREATE DATABASE bookflow_week2;
```

Query OK, 1 row affected (0.01 sec)

```
mysql>
```

```
mysql> USE bookflow_week2;
```

Database changed

```
mysql> CREATE TABLE Books (
```

```
-> book_id INT PRIMARY KEY,
```

```
-> title VARCHAR(100) NOT NULL,
```

```
-> isbn VARCHAR(20) UNIQUE,
```

```
-> published_year INT CHECK (published_year < 2027)
```

```
-> );
```

Query OK, 0 rows affected (0.06 sec)

```
INSERT INTO Books (book_id, title, isbn, published_year) VALUES
```

```
-> (1, 'The Great Gatsby', '9780743273565', 1925),
```

```
-> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
```

```
-> (3, '1984', '9780451524935', 1949);
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM Books;
```

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949

```
mysql> CREATE TABLE Members (
  -> member_id INT PRIMARY KEY,
  -> full_name VARCHAR(100),
  -> email VARCHAR(100) UNIQUE
  -> );
```

Query OK, 0 rows affected (0.06 sec)

```
INSERT INTO Members (member_id, full_name, email) VALUES
  -> (101, 'John Smith', 'john.smith@email.com'),
  -> (102, 'Emma Wilson', 'emma.wilson@email.com'),
  -> (103, 'Michael Brown', 'michael.brown@email.com');
```

Query OK, 3 rows affected (0.01 sec)

Records: 3 Duplicates: 0 Warnings: 0

```
SELECT * FROM Members;
```

member_id	full_name	email
-----------	-----------	-------

```

+-----+-----+-----+
| 101 | John Smith | john.smith@email.com |
| 102 | Emma Wilson | emma.wilson@email.com |
| 103 | Michael Brown | michael.brown@email.com |
+-----+-----+-----+

```

```

CREATE TABLE Loans (
  -> loan_id INT PRIMARY KEY,
  -> member_id INT,
  -> book_id INT,
  -> loan_date DATE,
  -> FOREIGN KEY (member_id) REFERENCES Members(member_id),
  -> FOREIGN KEY (book_id) REFERENCES Books(book_id)
  -> );

```

Query OK, 0 rows affected (0.08 sec)

```

INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
(1, 101, 1, '2025-01-05'),
(2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'),
(4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'),
(6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'),
(8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'),
(10, 101, 1, '2025-04-01');

```

```

SELECT * FROM Loans;

```

```

SELECT m.full_name AS Member_Name, b.title AS Book_Title

```

```

-> FROM Loans l
-> INNER JOIN Members m ON l.member_id = m.member_id
-> INNER JOIN Books b ON l.book_id = b.book_id;

```

Empty set (0.00 sec)

```

SELECT published_year, COUNT(book_id) AS Total_Books

```

```

-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;

```

```

+-----+-----+
| published_year | Total_Books |
+-----+-----+
|      1925    |      1    |
|      1949    |      1    |
|      1960    |      1    |
+-----+-----+

```

3 rows in set (0.01 sec)

```

SELECT published_year, COUNT(book_id) AS Total_Books

```

```

-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;

```

```

+-----+-----+
| published_year | Total_Books |
+-----+-----+
|      1925    |      1    |
|      1949    |      1    |
|      1960    |      1    |
+-----+-----+

```

3 rows in set (0.00 sec)

```
mysql> CREATE TABLE Donation_History (  
-> donation_id INT PRIMARY KEY,  
-> book_id INT,  
-> donor_name VARCHAR(100),  
-> donation_date DATE,  
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)  
-> );
```

Query OK, 0 rows affected (0.05 sec)

```
CREATE TABLE Donation_History (  
-> donation_id INT PRIMARY KEY,  
-> book_id INT,  
-> donor_name VARCHAR(100),  
-> donation_date DATE,  
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)  
-> );
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> INSERT INTO Books (book_id, title, isbn, published_year)  
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
```

Query OK, 1 row affected (0.00 sec)

```
mysql>
```

```
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name,  
donation_date)  
-> VALUES (1, 4, 'Raj Kumar', CURDATE());
```

Query OK, 1 row affected (0.00 sec)

```
mysql>
```

```
mysql> COMMIT;
```

Query OK, 0 rows affected (0.01 sec)

CREATE INDEX idx_books_isbn ON Books(isbn);

Query OK, 0 rows affected (0.05 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql>

mysql> SELECT * FROM Books

-> WHERE isbn = '9780451524935';

book_id	title	isbn	published_year
3	1984	9780451524935	1949

1 row in set (0.00 sec)

SELECT * FROM Books;

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
4	Animal Farm	9780451526342	1945

4 rows in set (0.00 sec)

mysql> SELECT * FROM Members;

member_id	full_name	email
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com

```
| 103 | Michael Brown | michael.brown@email.com |
```

```
+-----+-----+-----+-----+
```

3 rows in set (0.00 sec)

```
SELECT * FROM Donation_History;
```

```
mysql> SELECT * FROM Donation_History;
```

```
+-----+-----+-----+-----+
```

```
| donation_id | book_id | donor_name | donation_date |
```

```
+-----+-----+-----+-----+
```

```
| 1 | 4 | Raj Kumar | 2026-08-02 |
```

```
+-----+-----+-----+-----+
```

1 row in set (0.00 sec)INSERT INTO Loans (loan_id, member_id, book_id,
loan_date) VALUES

-> (1, 101, 1, '2025-01-05'),

-> (2, 102, 2, '2025-01-08'),

-> (3, 103, 3, '2025-01-10'),

-> (4, 101, 2, '2025-02-01'),

-> (5, 102, 1, '2025-02-05'),

-> (6, 103, 2, '2025-02-12'),

-> (7, 101, 3, '2025-03-01'),

-> (8, 102, 3, '2025-03-07'),

-> (9, 103, 1, '2025-03-15'),

-> (10, 101, 1, '2025-04-01');

Query OK, 10 rows affected (0.01 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
SELECT * FROM Loans;
```

```
+-----+-----+-----+-----+
```

```
| loan_id | member_id | book_id | loan_date |
```

```
+-----+-----+-----+-----+
```

```
| 1 | 101 | 1 | 2025-01-05 |
```

```
| 2 | 102 | 2 | 2025-01-08 |
```

	3		103		3		2025-01-10	
	4		101		2		2025-02-01	
	5		102		1		2025-02-05	
	6		103		2		2025-02-12	
	7		101		3		2025-03-01	
	8		102		3		2025-03-07	
	9		103		1		2025-03-15	
	10		101		1		2025-04-01	

+-----+-----+-----+-----+

10 rows in set (0.00 sec)